Rodrigo Grandy, PhD

Principal Scientist | Group Leader

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Summary

Innovative scientific leader with over 19 years of combined experience in academia and biotech, specialising in Cell and Molecular Biology with emphasis on Cell Therapy, RNA/cDNA technologies, and next-generation sequencing.

Currently developing an AI-driven biotech venture that leverages machine learning to design novel peptides and small molecules as cost-effective replacements for growth factors in cell-derived product manufacturing.

Certified in AI and Machine Learning for Data Science by the University of Cambridge, with expertise in neural networks, deep learning frameworks, LLMs, and RAG systems.

Skilled in establishing and managing R&D teams and projects across cell therapy and nucleic acid sequencing platforms, now enhanced by cutting-edge AI/ML capabilities for advanced data analysis and predictive modelling.

Strong combination of technical expertise spanning wet-lab methodologies and machine learning, with focus on data-driven decision-making, stakeholder management, and leveraging AI tools to unlock insights from complex datasets.

Core Expertise: Strategic R&D & team leadership • iPSC and hematopoietic stem cell systems • Genomics, bioinformatics & sequencing • Quality Control of iPSC systems • AI & ML • Neural networks & deep learning • LLMs & RAG systems • Data Analysis (NGS, Single-Cell RNA-Seq, Data Analysis & Interpretation)

Professional Experience

Evotec, United Kingdom | Germany (August 2023 - November 2024)
Principal Scientist, Project Lead, Line Manager: iPSC Cell Therapy/Immuno-Oncology

- Managed and supervised a team of scientists and technical staff to ensure the quality of work met company standards in iPSC-derived immunotherapies generation.
- Led scientific strategy and process optimisation for differentiating, expanding, and cryopreserving iPSC-derived HSCs and T cells.
- Contributed to strategic R&D planning: Identified strategies with the VP of immune oncology to control R&D expenses and FTEs within budget.
- Facilitated teamwork among various sites to improve Quality Control processes for iPSC maintenance, expansion, and MCB production.
- Coordinated inter-site actions to optimise the differentiation of hematopoietic stem cells from iPSCs.
- Spearheaded the creation of an mRNA-seq single cell atlas for internal use, focusing on immune cell differentiation during human embryonic development.
- Worked closely with the Business Development Director and SVP of Cell Therapy to showcase the company's iPSC-derived T cells program and cell therapy manufacturing capabilities to potential R&D and manufacturing corporate clients.
- Served as interim Line Manager for scientists leading iPSC-derived macrophage development, overseeing activities, setting quarterly objectives, and providing mentorship to enhance team performance.

- Partnered with senior leaders on key projects, including iPSC line generation in a GMP environment and establishing NGS platforms for assessing iPSC line genetic stability.

- Developed a comprehensive strategy for biomarker assay generation using flow cytometry to characterise cell types during T cell differentiation and intermediate stages.

- Supervised the characterisation of iPSC-derived T cell phenotypes and functionality, led by the immunology team in Toulouse, France.

Adaptimmune | Abingdon, UK (February 2022 - August 2023)
Group Leader/Line Manager Stem Cell Research

- Developed a quality control package in compliance with international standards to assess the quality of in-house derived iPS cell lines.

- Formed and led a team of senior scientists and scientists to manage the quality control program for our engineered iPS cells and derivatives.

- Conducted performance reviews and addressed any misalignment of team members with group objectives.

- Reported progress to stakeholders. Collaborated with project leads and other departments to ensure timely delivery of results.

- Directed the research program, improving the efficiency and consistency of iPS cell cultures, genetically engineered cell lines, and HSC differentiation systems to meet company objectives.

- Created and implemented a training program to standardise iPS cell culture techniques and ensure adherence to international guidelines for scientists.

- Coordinated the quality control, banking, and differentiation processes for hiPSC lines and HSC derivatives, facilitating the generation of allogenic T cells.

- Served as a subject matter expert on stem cell-related matters, providing strategic guidance and advice across organisation.

- Defined and managed the research program budget, ensuring financial resources were aligned with project goals.

- Engaged in constructing and improving the platform to support the development of engineered iPSC-derived T cell therapies.

Cytiva, Cardiff, UK (October 2019 - February 2022)
Senior Scientist Research and Development: Genomics & Cellular Research

- Generated and analysed NGS data to evaluate performance and quality of various Cytiva products and prototypes.

- Operated and maintained Next Generation Sequencer. Collaborated with the head of R&D to implement Single Cell analysis laboratory capabilities.

- Developed a method for detecting COVID-19 viral RNA through "Direct RT-PCR," optimising the formulation for reliable and consistent detection in human nasopharyngeal swab samples.

- Provided advisory support to R&D and G&D leads, helping them understand NGS and Single Cell Technology data, applications, and requirements, while collaborating with leadership to establish a new Single Cell Technology portfolio.

- Partnered with R&D and G&D leadership to establish new Single Cell Technology lab, expanding company's capabilities.

- Collaborated with the Lead Chemist to develop streptavidin-conjugated magnetic beads for use in hybridisation capture assays, including Exome sequencing.

- Supervised two junior scientists on a nucleic acid isolation project and a scientist on the Single Cell Technology portfolio.

- Led the team for Single Cell, NGS, and New Isolation Kit projects, overseeing junior scientists and supporting M4 activities for the VIA Omics Bundle.

Education, Training, and Certifications

- AI and Machine Learning for Data Science Certification
University of Cambridge (UK) | 2025

- Postdoctoral Researcher
Stem Cell Institute, Cambridge University (UK) | 2015-2019

- Postdoctoral Researcher
UMass Medical School (USA) | 2011-2015

- PhD in Cell and Molecular Biology
University of Concepcion (Chile) | 2011

- Bachelor's Degree in Biochemistry
University of Concepcion (Chile) | 2004

Publications & Conferences

Full list of publications available on: https://www.ncbi.nlm.nih.gov/myncbi/1xWYmtc7fmqI9z/bibliography/public/